

NAME: _____ PERIOD: _____ DATE: _____

What Can Each Study Claim?

AP Statistics · Unit 1 · Topic 1.10 · B: 2026-09-28 · E: 2026-10-05

[STAR] TODAY'S PROBLEM

Suppose a student wants to know whether a new tutoring center raises test scores.

Plan A: Compare this year's test scores for students who choose to attend the center with scores for students who do not attend.

Plan B: Recruit 40 student volunteers, then use a coin flip to assign each volunteer either to attend the center or not to attend. Compare their later test scores.

Two plans for studying tutoring

Plan	Who?	Groups?	Response
A	Students	Choose	Scores
B	40 volunteers	Coin flip	Scores

FIRST TAKE — before the video (5 min)

Which plan gives stronger evidence that the center raises scores? Name the feature behind your choice.

[BOOK] SELECTION, ASSIGNMENT, AND SCOPE (keep on the page)

In an **observational study**, investigators observe groups without assigning treatments; association does not establish cause. In an **experiment**, treatments are assigned. **Random assignment** supports a causal comparison by tending to balance lurking variables. **Random selection** or other evidence that the participants represent a population supports generalization to that population. Assignment and selection answer different questions.

Finish after the video:

DOK 1 (a) Identify Plan A as an observational study or experiment, and do the same for Plan B. State why for each.

DOK 2 (b) Name one lurking variable that could make Plan A's groups differ, and explain how it could be related to both attendance and test score.

DOK 3 (c) Which plan could support the claim that the center raises scores? Cite the design feature that settles it and explain what Plan A's result could mean instead. Then state one conclusion Plan B still could not support about students beyond the volunteers.

Frame: Plan _____ supports a causal claim because _____.

Frame: It still cannot generalize to _____ because _____.

EXIT — turn this in

One thing the video changed about my first take: _____